Experiment 1:

Problem Statement

A disease prediction system produced the following results.

```
Actual:
[1,1,1,0,0,0,1,0,1,0,1,0]

Predicted:
[1,0,1,0,0,1,1,0,0,0,1,0]
```

Calculate

- Precision
- Recall
- F1-score

Display the confusion matrix.

Experiment 2

Aim

Evaluate a spam detection classifier.

Problem Statement

Given

```
Actual

Spam
Spam
Spam
Not Spam
Spam
Not Spam
Spam
Not Spam
Spam
Not Spam
```

```
Predicted

Spam
Spam
Not Spam
Not Spam
Spam
Spam
Spam
Not Spam
Spam
Not Spam
```

Convert the labels into binary values and calculate

- Precision
- Recall
- F1-score

Experiment-3

Aim

Generate a complete classification report.

Problem Statement

1. Load the Iris dataset.
2. Split the dataset into training and testing sets.
3. Train a Decision Tree classifier.
4. Predict the test data.
5. Calculate

- Precision
- Recall
- F1-score
- Accuracy

using Scikit-learn.

Experiment 4:

Aim

Calculate regression error metrics for temperature prediction.

Problem Statement

Given the daily temperatures (°C):

```
Actual:
[30, 32, 31, 35, 36, 34, 33]

Predicted:
[29, 33, 30, 34, 37, 35, 32]
```

Write a Python program to compute:

- MAE
- MSE
- RMSE

Experiment 5

Aim

Evaluate the performance of a regression model using MAE, MSE, and RMSE.

Problem Statement

A regression model predicts students' final examination marks.

```
Actual Marks:
[78, 85, 90, 72, 88, 95, 80]

Predicted Marks:
[80, 82, 89, 75, 86, 93, 79]
```

Write a Python program to calculate:

- MAE
- MSE
- RMSE

Interpret the obtained error values.